LS problem: $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ $\quad x_i \in \mathbb{R}^n, \quad y_i \in \mathbb{R}$

$\Phi \leftarrow$ data $\quad \{f_1, \cdots, f_k\}$

$\Phi \in \mathbb{R}^{N \times k}$, y

$$\alpha^* = \underset{\alpha \in \mathbb{R}^k}{\operatorname{argmin}} \|\Phi\alpha - y\|_2^2 \iff$$

α^* satisfies normal eqn:

$$\Phi^T \Phi \alpha = \Phi^T y$$

: Gaussian elimination
(LU, $A = \Phi^T\Phi$
Cholesky)

: QR decomposition.
of $\Phi^T\Phi = QR$

$$QR\alpha = \Phi^T y$$

$$R\alpha = Q^T\Phi^T y$$

$\Phi \in \mathbb{R}^{N \times k}$ ← rectangular

$$\Phi = \begin{bmatrix} | & & | \\ \phi_1 & \cdots & \phi_k \\ | & & | \end{bmatrix}_{N \times k}$$

$N >> k$

Φ is full column rank.

$\text{rank}(\Phi) = k$

$\{\phi_1, \phi_2, \cdots, \phi_k\}$ is a linearly indep. set

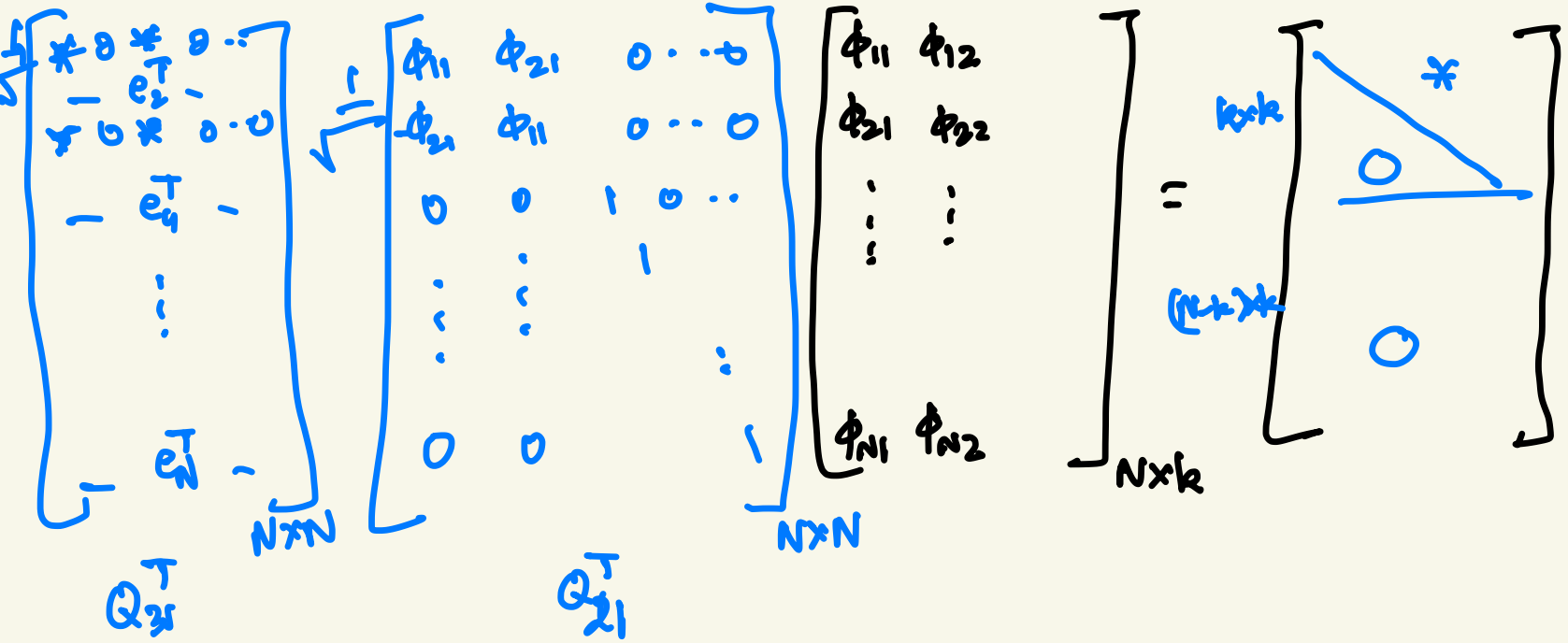
Find $\{q_1, q_2, \cdots, q_k\}$ orthonormal set such that

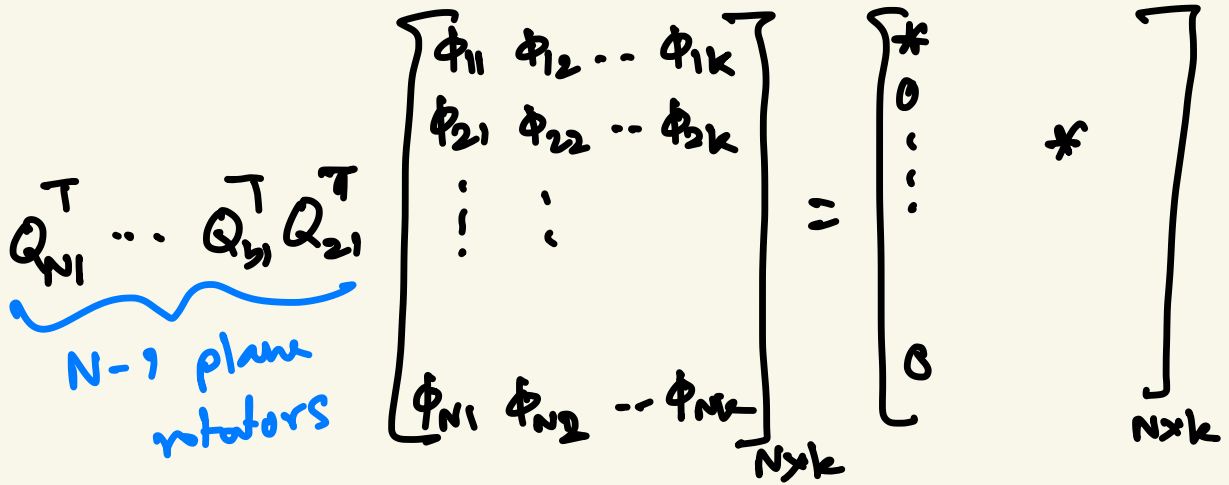
$\text{span}\{\phi_1\} = \text{span}\{q_1\}$

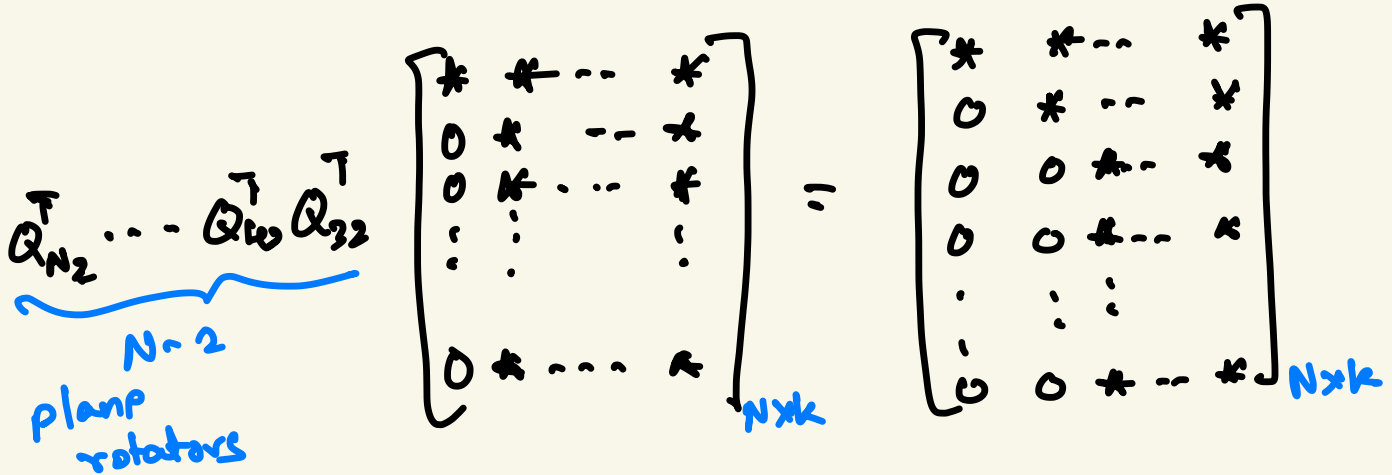
$\text{span}\{\phi_1, \phi_2\} = \text{span}\{q_1, q_2\}$

$\vdots$

$\text{span}\{\phi_1, \cdots, \phi_m\} = \text{span}\{q_1, q_2, \cdots, q_m\}$


N×N
N×N
N×k
k×k
(N-k)×k
Q_{2k}^T
Q_{21}^T


$Q_{N1}^T \cdots Q_{31}^T Q_{21}^T$
N-1 plane rotators
$\begin{bmatrix} \phi_{11} & \phi_{12} & \cdots & \phi_{1k} \\ \phi_{21} & \phi_{22} & \cdots & \phi_{2k} \\ \vdots & \vdots & & \\ \phi_{N1} & \phi_{N2} & \cdots & \phi_{Nk} \end{bmatrix}_{N\times k}$
$=$
$\begin{bmatrix} * \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{N\times k}$
$*$


$Q_{N2}^T \cdots Q_{42}^T Q_{32}^T$
N-2 plane rotators
$\begin{bmatrix} * & * & \cdots & * \\ 0 & * & \cdots & * \\ 0 & * & \cdots & * \\ \vdots & \vdots & & \vdots \\ 0 & * & \cdots & * \end{bmatrix}_{N\times k}$
$=$
$\begin{bmatrix} * & * & \cdots & * \\ 0 & * & \cdots & * \\ 0 & 0 & * & * \\ 0 & 0 & * & * \\ \vdots & \vdots & \vdots & \\ 0 & 0 & * & * \end{bmatrix}_{N\times k}$

$$Q^T \underline{\Phi} = \begin{bmatrix} * & * & * \\ & * & \\ & & * \\ & 0 & \\ \hline & 0 & \end{bmatrix} \left.\begin{matrix} \\ \\ \\ \\ \\ \end{matrix}\right\} k \times k \text{ square}$$

$$\Rightarrow \underset{N\times k}{\underline{\Phi}} = \underset{N\times N}{Q} \begin{bmatrix} R \\ \hline 0 \end{bmatrix}_{N\times k} \left.\begin{matrix} \\ \\ \end{matrix}\right\} k \times k$$

and $Q \in \mathbb{R}^{N\times N}$ orthogonal matrix

& $R \in \mathbb{R}^{k\times k}$ upper triangular matrix.

$$\Phi = Q \begin{bmatrix} R \\ 0 \end{bmatrix}$$ where $\Phi \in \mathbb{R}^{N \times k}$ full column rank matrix

$Q \in \mathbb{R}^{N \times N}$ orthogonal matrix

$R \in \mathbb{R}^{k \times k}$ is an invertible upper triangular matrix.

QR decomposition of $\Phi \in \mathbb{R}^{N \times k}$

Recall: $Q \in \mathbb{R}^{N \times N}$ orthogonal matrix, $z \in \mathbb{R}^N$

$$\|z\|_2^2 = \|Qz\|_2^2 = (Qz)^T Qz$$

$$\|z\|_2^2 = z^T z$$

$$= z^T (Q^T Q) z$$

$$= z^T z$$

Cost function: of LS problem

$\Phi \in \mathbb{R}^{N \times k}$ $\alpha \in \mathbb{R}^{k \times 1}$ $y \in \mathbb{R}^{N \times 1}$

$$\| \Phi\alpha - y \|_2^2$$

$$= \| Q^T(\Phi\alpha - y) \|_2^2$$

$$= \| Q^T\Phi\alpha - Q^T y \|_2^2$$

$$= \left\| \begin{bmatrix} R \\ 0 \end{bmatrix}\alpha - Q^T y \right\|_2^2$$

($\begin{bmatrix} R \\ 0 \end{bmatrix}\alpha$: $N \times 1$; $Q^T y$: $N \times 1$)

$$= \left\| \begin{bmatrix} R\alpha \\ 0 \end{bmatrix} - \begin{bmatrix} b \\ c \end{bmatrix} \right\|_2^2$$

($R\alpha$: $k \times 1$; 0: $N-k$; b: $k \times 1$; c: $N-k$)

$$= \left\| \begin{bmatrix} R\alpha - b \\ -c \end{bmatrix} \right\|_2^2$$

$$= \|R\alpha - b\|_2^2 + \|c\|_2^2$$

where $Q \in \mathbb{R}^{N \times N}$ is orthogonal s.t.

$$\Phi = Q\begin{bmatrix} R \\ 0 \end{bmatrix}$$

$$\Rightarrow Q^T\Phi = \begin{bmatrix} R \\ 0 \end{bmatrix} \begin{matrix} \} k \times k \\ \} (N-k) \times k \end{matrix}$$

$$\|\Phi\alpha - y\|_2^2 = \|R\alpha - b\|_2^2 + \|c\|_2^2 \qquad Q^T y = \begin{bmatrix} b \\ \hline c \end{bmatrix} \begin{matrix} \leftarrow k\times 1 \\ \leftarrow (N-k)\times 1 \end{matrix}$$

$$\Rightarrow \min_{\alpha \in \mathbb{R}^k} \|\Phi\alpha - y\|_2^2 = \min_{\alpha \in \mathbb{R}^k} \left[\|R\alpha - b\|_2^2 + \|c\|_2^2 \right]$$

$$\|R\alpha - b\|_2^2 + \|c\|_2^2 \geq \|c\|_2^2$$

Can I choose $\alpha \in \mathbb{R}^k$ s.t. $\|R\alpha - b\|_2^2 + \|c\|_2^2 = \|c\|_2^2$

$$R\alpha = b \Rightarrow \begin{bmatrix} * & * & & \\ 0 & \ddots & & \\ & & * & \end{bmatrix} \alpha = b$$

$$\min_{\alpha \in \mathbb{R}^k} \| \underline{\Phi}\alpha - y \|_2^2$$

$$Q^T \underline{\Phi} = \begin{bmatrix} * & \cdots & * \\ & \ddots & \vdots \\ 0 & & * \\ \hline 0 & & \end{bmatrix} \} R \quad k \times k \text{ upper triangular}$$

$$Q^T y = \begin{bmatrix} b \\ \hline c \\ \\ \end{bmatrix} \begin{matrix} \} k \times 1 \\ \} (N-k) \times 1 \\ N \times 1 \end{matrix}$$

Solve: $R\alpha = b$ by backward substitution

Note that QR decomposition $\underline{\Phi} \in \mathbb{R}^{N \times k}$ avoids calculating $\underline{\Phi}^T \underline{\Phi}$ and then solving normal-eqn's.

In general, (numerically) $\underline{\Phi}^T \underline{\Phi}$ is not recommended.

Consider $\Phi = \begin{bmatrix} 1 & 1 \\ 0 & \varepsilon \\ 1 & 1+\varepsilon^2 \end{bmatrix}$

$\Phi^T\Phi = \begin{bmatrix} 2 & 2+\varepsilon \\ 2+\varepsilon & 2+\varepsilon+2\varepsilon^2 \end{bmatrix}$

$\Phi^T\Phi = \begin{bmatrix} 2+\varepsilon+\varepsilon^2 & 2+\varepsilon+\varepsilon^2 \\ 2+\varepsilon+\varepsilon^2 & 2+\varepsilon+\varepsilon^4 \end{bmatrix}$

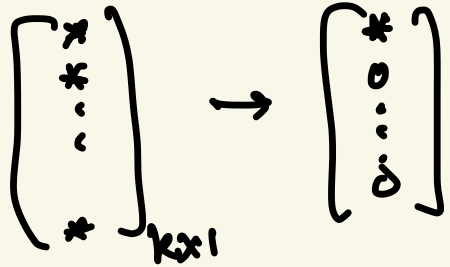
High condition number.

$\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ \sqrt{3+\varepsilon} & \cdot \end{bmatrix}$

QR - decomposition

→ rotators

→ reflectors



in one-shot